Discrete Random Variables and their distribution

In this chapter, we combine the methods of descriptive statistics and those of probability to describe and analyze probability distributions. Probability distributions describe what will probably happen instead of what actually did happen, and they are often given in the format of a graph, table, or formula.

Roll a die

Collect sample data, then get statistics and graphs.

x	f
1	8
2	10
3	9
4	12
5	11
6	10

$\bar{x} = 3.6$
$s = 1.7$

Find the probability for each outcome.

$P(1) = 1/6$
$P(2) = 1/6$
$\vdots$
$P(6) = 1/6$

Create a theoretical model describing how the experiment is expected to behave, then get its parameters.

x	$P(x)$
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

$\mu = 3.5$
$\sigma = 1.7$

DEFINITION 3.1

A **random variable** is a function of an outcome,

$$X = f(\omega).$$

In other words, it is a quantity that depends on chance.

The domain of a random variable is the sample space Ω. Its range can be the set of all real numbers $\mathbb{R}$, or only the positive numbers $(0, +\infty)$, or the integers $\mathbb{Z}$, or the interval $(0, 1)$, etc., depending on what possible values the random variable can potentially take.

Once an experiment is completed, and the outcome ω is known, the value of random variable $X(\omega)$ becomes determined.

Example:

Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. The possible outcomes and the values y of the random variable Y, where Y is the number of red balls, are

Sample Space	y
RR	2
RB	1
BR	1
BB	0

A random variable that takes on a finite or countably infinite number of values is called a **Discrete Random Variable**.

A random variable that takes on a non-countable, infinite number of values is a **Continuous Random Variable**.

Example 3.1. Consider an experiment of tossing 3 fair coins and counting the number of heads. Certainly, the same model suits the number of girls in a family with 3 children, the number of 1's in a random binary string of 3 characters, etc.

Let X be the number of heads (girls, 1's). Prior to an experiment, its value is not known. All we can say is that X has to be an integer between 0 and 3. Since assuming each value is an event, we can compute probabilities,

$$\begin{aligned}
\boldsymbol{P}\{X=0\} &= \boldsymbol{P}\{\text{three tails}\} = \boldsymbol{P}\{TTT\} = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{8} \\
\boldsymbol{P}\{X=1\} &= \boldsymbol{P}\{HTT\} + \boldsymbol{P}\{THT\} + \boldsymbol{P}\{TTH\} = \frac{3}{8} \\
\boldsymbol{P}\{X=2\} &= \boldsymbol{P}\{HHT\} + \boldsymbol{P}\{HTH\} + \boldsymbol{P}\{THH\} = \frac{3}{8} \\
\boldsymbol{P}\{X=3\} &= \boldsymbol{P}\{HHH\} = \frac{1}{8}
\end{aligned}$$

Summarizing,

x	$\boldsymbol{P}\{X = x\}$
0	$1/8$
1	$3/8$
2	$3/8$
3	$1/8$
Total	1

$\Diamond$

- Notation:
  - random variable X
  - numerical value x

DEFINITION 3.2

Collection of all the probabilities related to X is the **distribution** of X. The function

$$P(x) = \boldsymbol{P}\{X = x\}$$

is the **probability mass function**, or **pmf**. The **cumulative distribution function**, or **cdf** is defined as

$$F(x) = \boldsymbol{P}\{X \le x\} = \sum_{y \le x} \boldsymbol{P}(y). \tag{3.1}$$

The set of possible values of X is called the **support** of the distribution F.

Looking at (3.1), we can conclude that the cdf $F(x)$ is a non-decreasing function of x, always between 0 and 1, with

$$\lim_{x \downarrow -\infty} F(x) = 0 \text{ and } \lim_{x \uparrow +\infty} F(x) = 1.$$

When A is an interval, its probability can be computed directly from the cdf $F(x)$,

$$\boldsymbol{P}\{a < X \leq b\} = F(b) - F(a).$$

Probability Mass Function:

The set of ordered pairs $(x, f(x))$ is a **probability function**, **probability mass function**, or **probability distribution** of the discrete random variable X if, for each possible outcome x,

1. $f(x) \geq 0$,
2. $\sum\limits_{x} f(x) = 1$,
3. $P(X = x) = f(x)$.

3.11 A shipment of 7 television sets contains 2 defective sets. A hotel makes a random purchase of 3 of the sets. If x is the number of defective sets purchased by the hotel, find the probability distribution of X. Express the results graphically as a probability histogram.

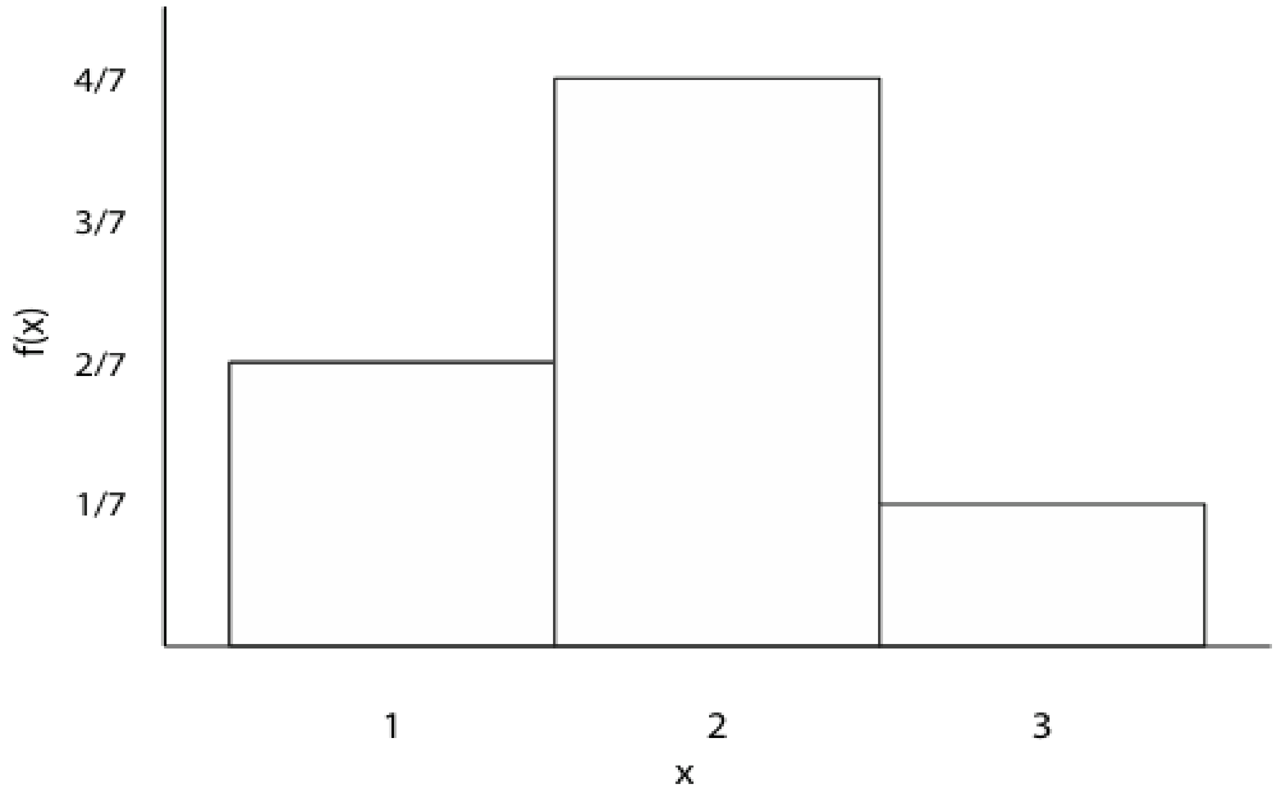
We can select x defective sets from 2, and $3-x$ good sets from 5 in $\binom{2}{x}\binom{5}{3-x}$ ways. A random selection of 3 from 7 sets can be made in $\binom{7}{3}$ ways. Therefore,

$$f(x) = \frac{\binom{2}{x}\binom{5}{3-x}}{\binom{7}{3}}, \qquad x = 0, 1, 2.$$

In tabular form

x	0	1	2
$f(x)$	2/7	4/7	1/7

The following is a probability histogram:



3.15 Find the cumulative distribution function of the random variable X representing the number of defectives in Exercise 3.11. Then using $F(x)$, find

(a) $P(X = 1)$;

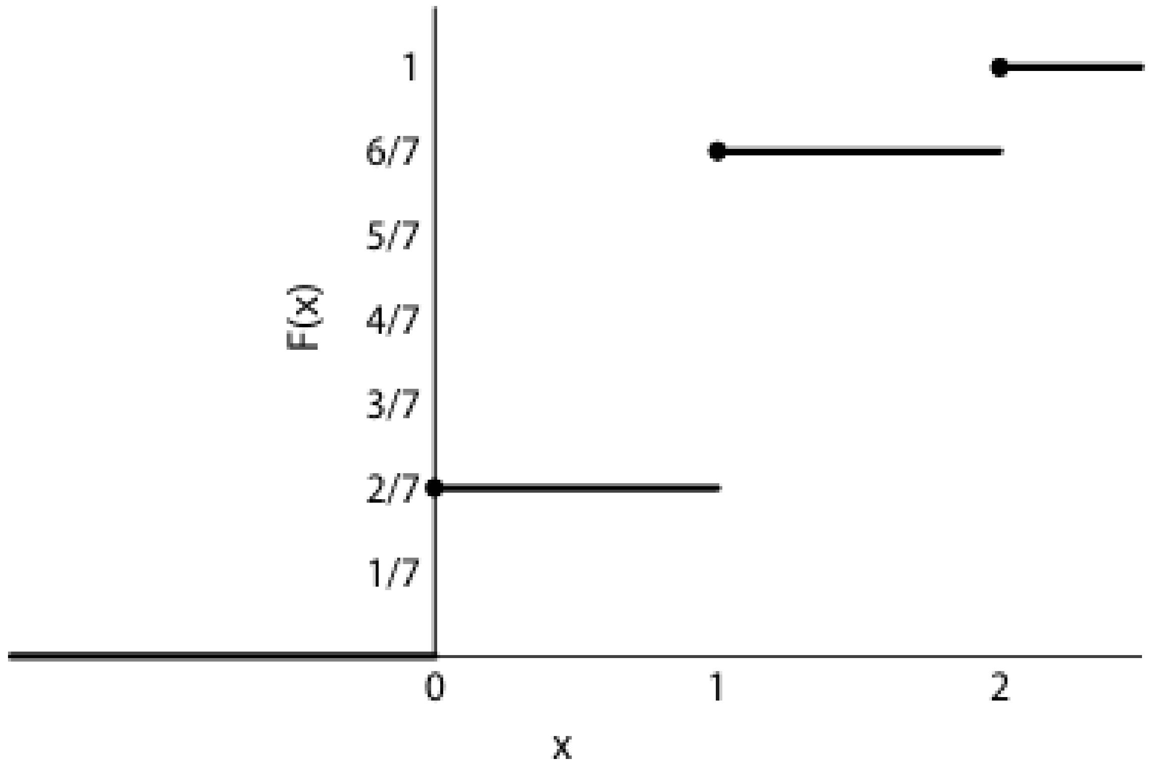
(b) $P(0 < X \leq 2)$.

3.15 The c.d.f. of X is

$$F(x) = \begin{cases} 0, & \text{for } x < 0, \\ 2/7, & \text{for } 0 \leq x < 1, \\ 6/7, & \text{for } 1 \leq x < 2, \\ 1, & \text{for } x \geq 2. \end{cases}$$

(a) $P(X = 1) = P(X \leq 1) - P(X \leq 0) = 6/7 - 2/7 = 4/7$;

(b) $P(0 < X \leq 2) = P(X \leq 2) - P(X \leq 0) = 1 - 2/7 = 5/7$.



1
6/7
5/7
F(x)
4/7
3/7
2/7
1/7
0
1
2
x

Example 3.3 (ERRORS IN INDEPENDENT MODULES). A program consists of two modules. The number of errors X_1 in the first module has the pmf $P_1(x)$, and the number of errors X_2 in the second module has the pmf $P_2(x)$, independently of X_1, where

x	$P_1(x)$	$P_2(x)$
0	0.5	0.7
1	0.3	0.2
2	0.1	0.1
3	0.1	0

Find the pmf and cdf of $Y = X_1 + X_2$, the total number of errors.

Solution. We break the problem into steps. First, determine all possible values of Y, then compute the probability of each value. Clearly, the number of errors Y is an integer that can be as low as $0+0=0$ and as high as $3+2=5$. Since $P_2(3)=0$, the second module has at most 2 errors. Next,

$$
\begin{aligned}
P_Y(0) &= P\{Y=0\} &&= \boldsymbol{P}\{X_1=X_2=0\} = P_1(0)P_2(0) \\
&&&= (0.5)(0.7) = 0.35 \\
P_Y(1) &= P\{Y=1\} &&= P_1(0)P_2(1)+P_1(1)P_2(0) \\
&&&= (0.5)(0.2)+(0.3)(0.7) = 0.31 \\
P_Y(2) &= P\{Y=2\} &&= P_1(0)P_2(2)+P_1(1)P_2(1)+P_1(2)P_2(0) \\
&&&= (0.5)(0.1)+(0.3)(0.2)+(0.1)(0.7) = 0.18 \\
P_Y(3) &= P\{Y=3\} &&= P_1(1)P_2(2)+P_1(2)P_2(1)+P_1(3)P_2(0) \\
&&&= (0.3)(0.1)+(0.1)(0.2)+(0.1)(0.7) = 0.12 \\
P_Y(4) &= P\{Y=4\} &&= P_1(2)P_2(2)+P_1(3)P_2(1) \\
&&&= (0.1)(0.1)+(0.1)(0.2) = 0.03 \\
P_Y(5) &= P\{Y=5\} &&= P_1(3)P_2(2) = (0.1)(0.1) = 0.01
\end{aligned}
$$

Now check:

$$\sum_{y=0}^{5} P_Y(y) = 0.35+0.31+0.18+0.12+0.03+0.01 = 1,$$

thus we *probably* counted all the possibilities and did not miss any. (We just wanted to emphasize that simply getting $\sum P(x) = 1$ does not guarantee that we made no mistake in our solution. However, if this equality is not satisfied, we have a mistake for sure.)

The cumulative function can be computed as

$$\begin{aligned}
F_Y(0) &= P_Y(0) = 0.35 \\
F_Y(1) &= F_Y(0) + P_Y(1) = 0.35 + 0.31 = 0.66 \\
F_Y(2) &= F_Y(1) + P_Y(2) = 0.66 + 0.18 = 0.84 \\
F_Y(3) &= F_Y(2) + P_Y(3) = 0.84 + 0.12 = 0.96 \\
F_Y(4) &= F_Y(3) + P_Y(4) = 0.96 + 0.03 = 0.99 \\
F_Y(5) &= F_Y(4) + P_Y(5) = 0.99 + 0.01 = 1.00
\end{aligned}$$

Between the values of Y, $F(x)$ is constant. $\Diamond$